

RAY POMPONIO

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PROFESSIONAL SUMMARY

Data Scientist with 5 years of experience in biomedical applications, specializing in statistical analysis and collaborative research. Proven ability to derive insights from complex datasets and contribute to impactful scientific publications. Seeking roles with opportunities for leadership in research and healthcare analytics.

PROJECT EXPERIENCE

Data Quality Overhaul for CDC-Sponsored Surveillance Study

July 2024 – December 2024

- Co-led initiative to revise data quality workflow in study with 3,000+ annual enrollees.
- Resulted in resolution of over 2,500 flags catching missing or mis-entered data.
- Overall reduction in CDC-prompted queries of local database, previously thousands of queries were triggered and now fewer than 200 monthly queries are anticipated.

Dashboard Design in Power BI

November 2024 – January 2025

- Created dashboards to monitor live enrollment and track virus circulation patterns in pediatric research study with 100+ laboratory results per week.

Electronic Consent Form Creation and Validation

August 2024 – Present

- Utilizing REDCap's e-consent framework, 25+ critical errors were flagged by automated detection.

WORK EXPERIENCE

Data Scientist

University of Pittsburgh, Department of Pediatrics – Pittsburgh, PA

May 2024 – Present

- Conduct statistical analyses with biomedical datasets to support pediatric research initiatives.
- Collaborate with cross-functional clinical teams to design studies and interpret results.
- Develop reproducible data pipelines and dashboards for ongoing studies.
- Present findings to stakeholders and contribute to publications.

Graduate Research Assistant

Colorado School of Public Health, Department of Biostatistics – Denver, CO

August 2021 – December 2023

- Led data analysis for public health research projects focused on pulmonology and health outcomes.
- Applied statistical methods and R/Python programming to clean, analyze, and visualize data.
- Co-authored peer-reviewed publications and presented at academic conferences.

Data Analyst

University of Pennsylvania, Department of Radiology – Philadelphia, PA

March 2018 – August 2020

- Analyzed imaging and clinical data to support research in radiology and neuroscience.
- Built Python software to streamline combination of diverse neuroimaging datasets.

- Collaborated with radiologists and researchers to identify lifespan trends and improve data quality.
- Contributed to the development of predictive models of age, dementia for clinical diagnostics.

EDUCATION

Master of Science in Biostatistics

University of Colorado Anschutz Medical Campus – Aurora, CO

Graduated: May 2023

Bachelor of Science in Economics

University of Pennsylvania, Wharton School – Philadelphia, PA

Graduated: May 2018

SKILLS

- Statistical Programming: R, SAS, Python
- Data Manipulation & Analysis: Regression, longitudinal modeling, survival analysis
- Scripting & Automation: Unix Bash, Microsoft Power Automate
- Data Visualization: ggplot2, shiny, matplotlib, Power BI
- Soft Skills: Collaborative research, scientific communication, problem-solving

SELECTED PUBLICATIONS

Pomponio, R., et al. (2020). Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *Neuroimage*, 208(1).

Habes, M., et al. (2020). The Brain Chart of Aging: Machine-learning analytics reveals links between brain aging, white matter disease, amyloid burden, and cognition in the iSTAGING consortium of 10,216 harmonized MR scans. *Alzheimer's & Dementia*, 17(1), 89-102.

Nasrallah, I., et al. (2021). Association of Intensive vs Standard Blood Pressure Control With Magnetic Resonance Imaging Biomarkers of Alzheimer Disease: Secondary Analysis of the SPRINT MIND Randomized Trial. *JAMA Neurology*, 78(5), 568-577.

Chen, A., et al. (2022). Harmonizing functional connectivity reduces scanner effects in community detection. *Neuroimage*, 256(1).

Dunbar, P., et al. (2024). Analgesia and Sedation Use During Noninvasive Ventilation for Acute Respiratory Failure. *Critical Care Medicine*, 52(7), 1043-1053.

Burnham, E., et al. (2024). Prevalence of Alcohol Use Characterized by Phosphatidylethanol in Patients With Respiratory Failure Before and During the COVID-19 Pandemic. *CHEST Critical Care*, 2(1).

Smith, J., et al. (2024). Donor derived cell free DNA in lung transplant recipients rises in setting of allograft instability. *Frontiers in Transplantation*, 3.

Held, N., et al. (2024). Extraneous load, patient census, and patient acuity correlate with cognitive load during ICU rounds. *Chest*, 165(6), 1448-1457.

Pomponio, R., et al. (2025). Phosphatidylethanol measures in patients with severe COVID-19-associated respiratory failure identify a subset with alcohol misuse. *Alcohol: Clinical & Experimental Research*, 49(1), 165-174.

OTHER RELEVANT EXPERIENCE

Peer reviewer for four journal articles (2020-present), Graduate teaching assistant (Fall 2023)

Maintainer for software package 'neuroHarmonize' (2021): available on Python Packaging Index (PyPI)